🔐 1️⃣ AUTHENTICATION & USER MANAGEMENT (HIGH VALUE)

Features

- Email + Password login
- Google OAuth (optional)
- User profile dashboard
- API key saved securely per user

Why it matters

- Enables **subscriptions**
- Tracks usage
- Prevents abuse

Tech

- NextAuth.js / Clerk (free tier)
- JWT (backend)

💳 2️⃣ USAGE LIMITS & SUBSCRIPTION LOGIC (💰 MONEY MAKER)

Features

Plan	Limit
Free	1 project / day
Pro	Unlimited
College	Bulk generation

Add

- Request counter
- Monthly reset
- Soft paywall UI

UI Ideas

- “You have 1 project left today”
- Upgrade modal

- Locked buttons 🔒

🧠 3️⃣ SMART VIVA ASSISTANT (EXAM READY)

Advanced Viva Modes

- 🎓 **Examiner Mode** (strict answers)
- 👩‍🎓 **Student Mode** (simple explanation)
- 🧪 **Follow-up Question Mode**
- 🧠 **Why? / How? / Compare with X**

Extra

- "Ask from my project only"
- "Explain this diagram"
- "Predict next viva question"

💥 This is a **killer differentiator**

📄 4️⃣ PROFESSIONAL DOCUMENT GENERATION

Report Enhancements

- College name & student name
- Roll number
- Guide name
- Semester
- IEEE / ACM format

PPT Enhancements

- Animated slides
- Architecture diagrams
- Flowcharts
- Auto speaker notes

📦 5️⃣ PROJECT ZIP EXPORT (INDUSTRY STYLE)

Generate a **real developer repo**:

```
project_name/
├── backend/
├── frontend/
├── dataset/
├── README.md
├── requirements.txt
```

Include:

- Setup instructions
- Output screenshots
- Test cases

🧠 6️⃣ PROJECT CUSTOMIZATION PANEL

Let users:

- Regenerate only **Abstract**
- Regenerate only **Viva**
- Increase complexity
- Convert project to:
  - Web app
  - Mobile app
  - Cloud-based

🧪 7️⃣ PLAGIARISM & UNIQUENESS CHECK (🔥 GOLD FEATURE)

Features

- AI-based similarity score
- Auto re-phrasing

- “Make it more unique”
- “Humanize report”

Very attractive for colleges 👀

⚡ 8️⃣ PERFORMANCE & UX POLISH

Frontend

- Skeleton loaders
- Streaming responses
- Progress bar (Abstract → Code → PPT)

Backend

- Task queue (Celery / BackgroundTasks)
- Request caching
- Timeout handling

📊 9️⃣ ANALYTICS DASHBOARD (ADMIN + USER)

User Dashboard

- Projects generated
- Viva questions practiced
- Weak topics

Admin Dashboard

- Popular domains
- Conversion rate
- Daily active users

🧩 🔟 INTEGRATIONS (OPTIONAL BUT IMPRESSIVE)

- GitHub repo creation
- Google Drive export
- PDF watermark removal (Pro)

- Resume bullet generator

🧠 FEATURE PRIORITY (DO THIS ORDER)

Priority	Feature
⭐⭐⭐⭐⭐	Auth + Usage Limits
⭐⭐⭐⭐⭐	Advanced Viva Assistant
⭐⭐⭐⭐	Pro PDF/PPT
⭐⭐⭐⭐	Project ZIP
⭐⭐⭐	Plagiarism
⭐⭐⭐	Analytics

🏆 HOW THIS LOOKS ON RESUME

AI Project Generator SaaS
Built a full-stack AI SaaS using Next.js, FastAPI, and LLMs to auto-generate academic projects, reports, PPTs, and viva assistance. Implemented authentication, subscription logic, document automation, and scalable API architecture.

🔥 **Recruiters will love this**